HDMI

MS2109
HDMI captrue
USB

RJ45

DP83825IRMQR
100M ETH PHY
RMII

USB-C
(Debug)

CH343p_0
external USB to UART
UART0, CHIP_PU, GPIO35
Debug

TF card
Storage
SDIO

2*Reserved GPIO
1*I2C, shared with MCU and MS2109

P4-core

ESP32-P4NRW32X
128Mbit SPI Flash

CAUTION — CHIP REVISION
ESP32-P4NRW16/32 and ESP32-P4NRW16X/32X use different IC revisions and are not drop-in compatible.

Non-X version, chip revision v1.x: Pin 54 is NC and must be left unconnected.

X version, chip revision v3.x: Pin 54 is VDD_HP and must be connected to the VDD_HP power rail.

v1.x and v3.x require separate firmware builds and cannot use the same firmware image.

ESP32-P4NRW16/32 is planned for EOL. ESP32-P4NRW16X/32X is recommended for new designs. But some Dev Board still use Non-X version chip

Verify the exact chip part number and revision before PCB assembly.

OC PWRcontrol
AQS221R2S
4_Channel GPIO

PowerTree
3V3
TLV62569DBVR

1V2
TLV62569DBVR

VCCHP(1V2)
TLV62569DBVR
Use MCU FB/EN

CH343p_1
external USB to UART
UART1

USB_HID
GPIO23, 22

8PIN
Power Control

9PIN USB

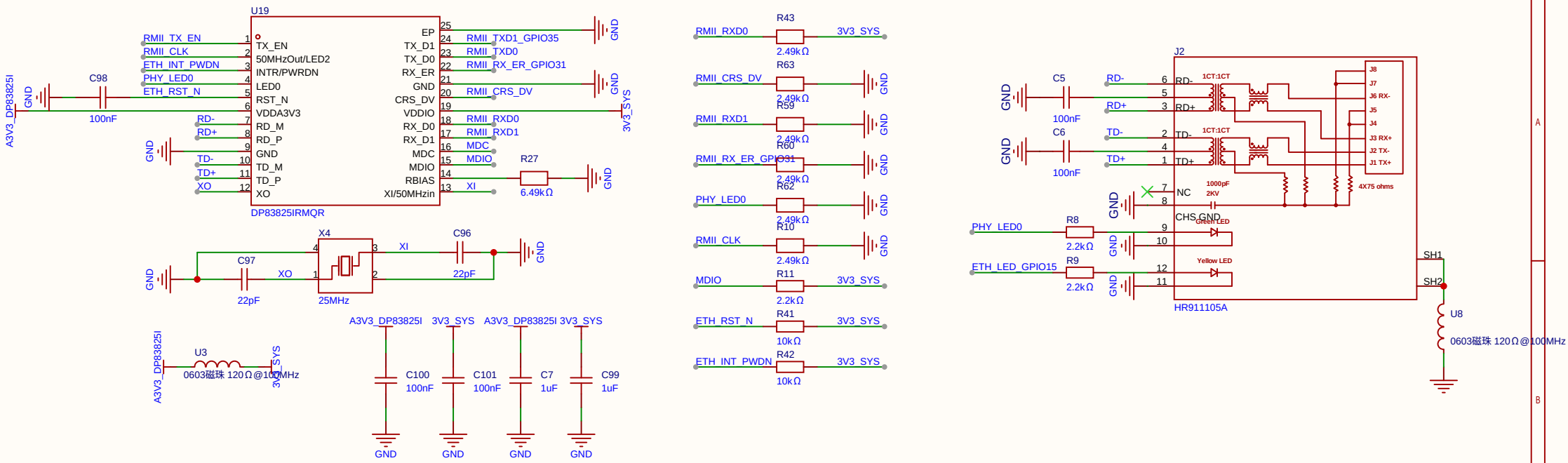
PCB Fabrication Recommendation

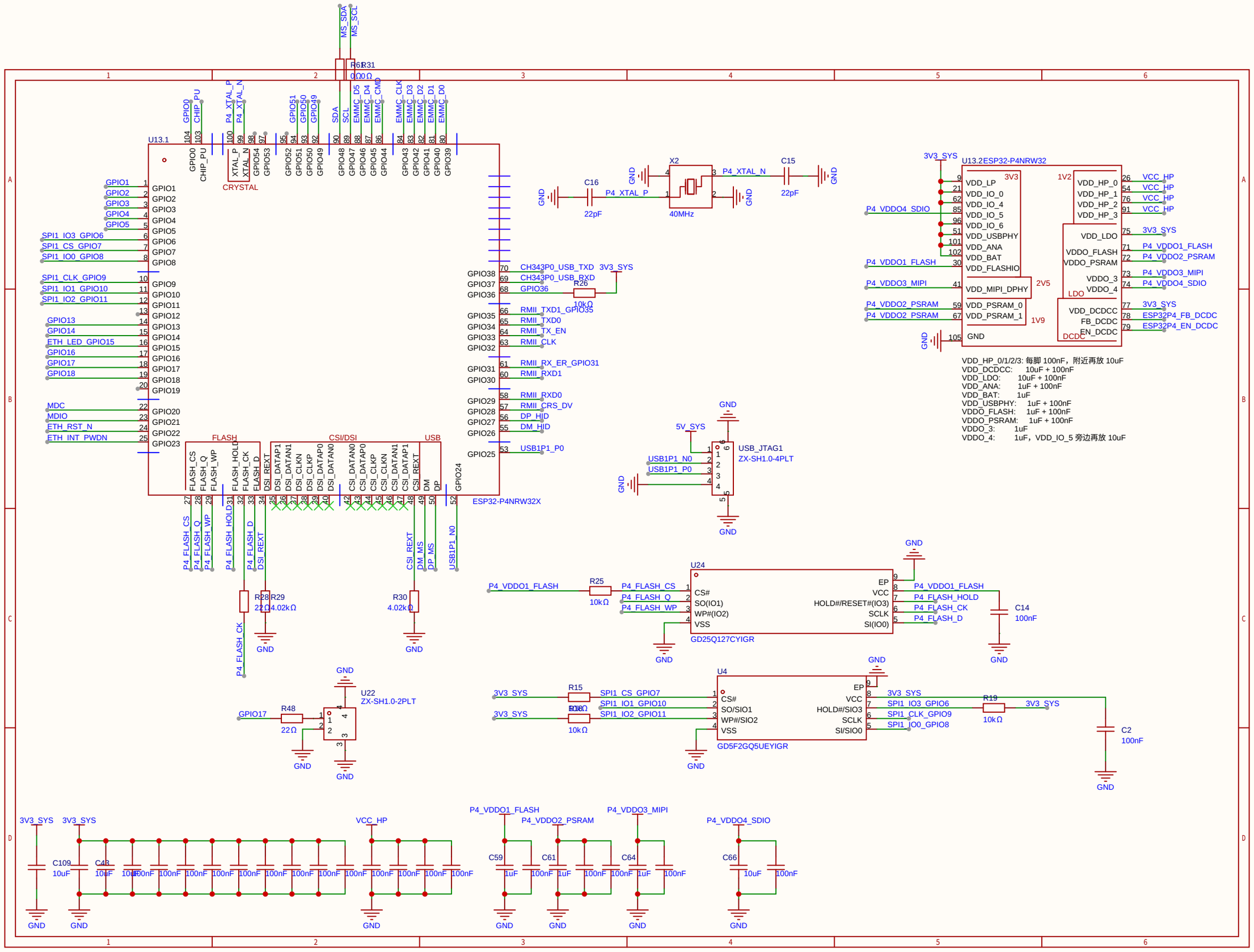
The current PCB is designed as a 4-layer board. Due to the impedance-control and reference-plane continuity requirements of the high-speed signals routed on the bottom layer, a 6-layer stack-up is recommended for prototype fabrication, with Layer 5 implemented as a complete and continuous ground plane.

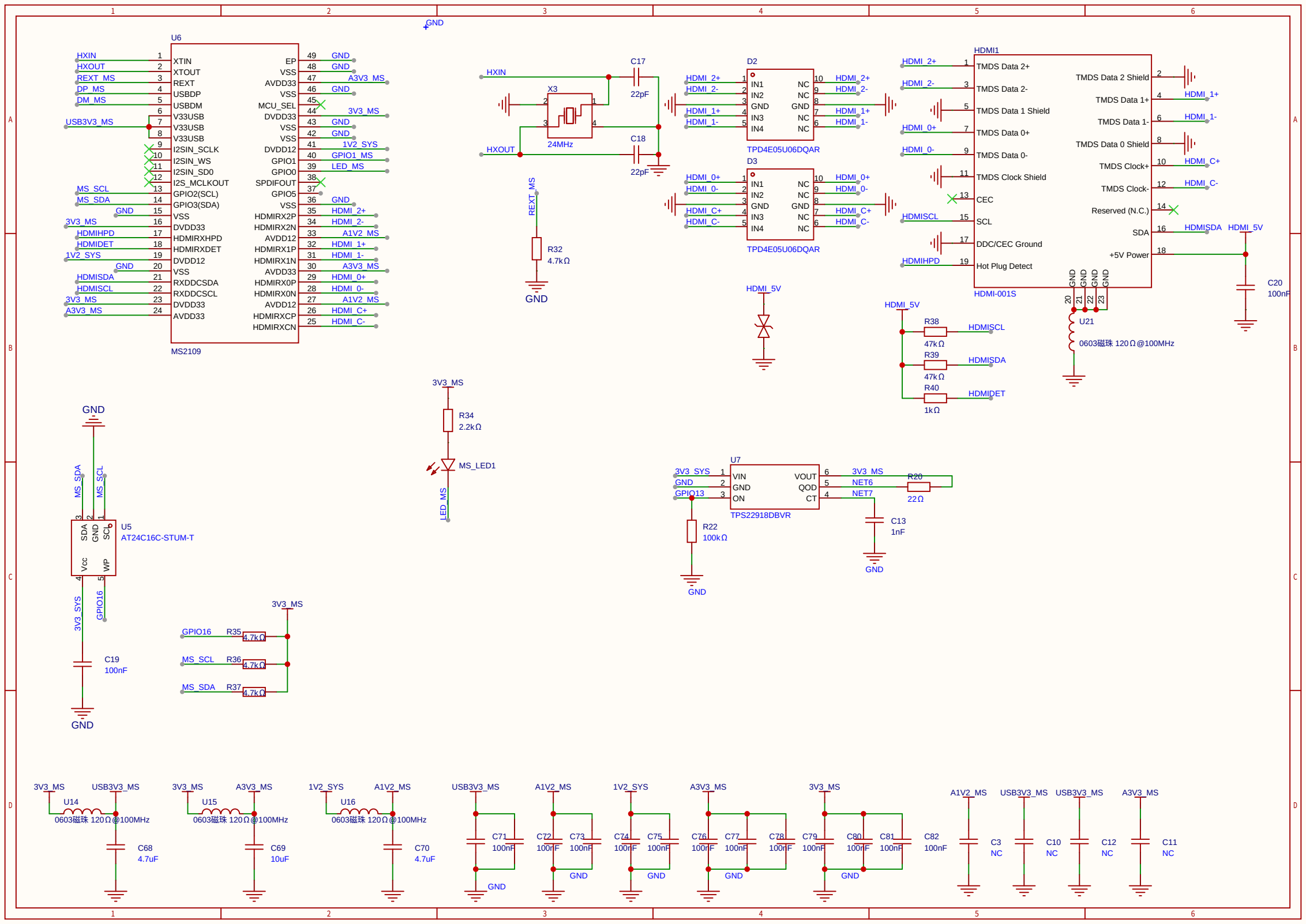
The solid Layer 5 ground plane provides a stable reference plane and return path for the bottom-layer high-speed signals, improving impedance matching and reducing signal-integrity and EMI risks. Before fabrication, confirm the final stack-up with the PCB manufacturer and recalculate the trace widths and spacing according to the specified target impedance.

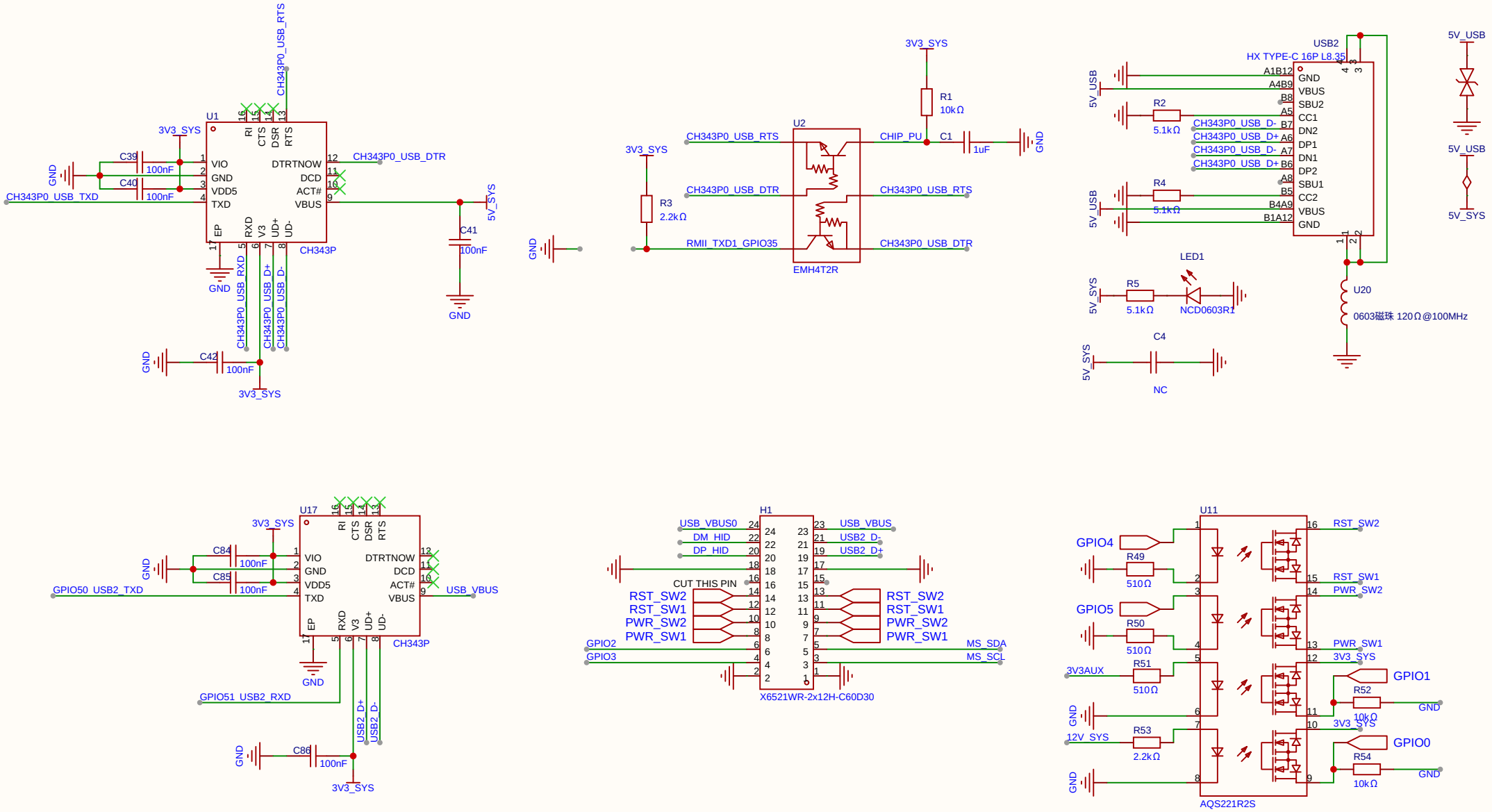
JLC04161H-3313 Impedance	Type	Layer	Width(mils)	Spacing(mils)
90Ω USB	Differential Pair	Outer	6.1	8
90Ω USB	Differential Pair	Inner	5.2	8
100Ω ETH, HDMI	Differential Pair	Outer	4.7	8
100Ω ETH, HDMI	Differential Pair	Inner	4.1	8
50Ω RMII, SDIO	Single Ended	Outer	5.9	8
50Ω RMII, SDIO	Single Ended	Inner	5.3	8

Typical Power Consumption
720P60 1.2W(Input)


U19
RMII_TX_EN
RMII_CLK
ETH_INT_PWDN
PHY_LED0
ETH_RST_N
TX_EN
50MHzOut/LED2
INTR/PWRDN
LED0
RST_N
VDDA3V3
RD_M
RD_P
GND
TD_M
TD_P
XO
EP
TX_D1
TX_D0
RX_ER
GND
CRS_DV
VDDIO
RX_D0
RX_D1
MDC
MDIO
RBIAS
XI/50MHzin
RMII_TXD1_GPIO35
RMII_TXD0
RMII_RX_ER_GPIO31
RMII_CRS_DV
RMII_RXD0
RMII_RXD1
MDC
MDIO
XI
DP83825IRMQR
C98
100nF
R27
6.49kΩ
X4
25MHz
C97
22pF
C96
22pF
U3
0603磁珠 120Ω@100MHz
C100
100nF
C101
100nF
C7
1uF
C99
1uF
R43
2.49kΩ
R63
2.49kΩ
R59
2.49kΩ
R60
2.49kΩ
R62
2.49kΩ
R10
2.49kΩ
R11
2.2kΩ
R41
10kΩ
R42
10kΩ
J2
C5
C6
100nF
1CT:1CT
1000pF
2KV
4X75 ohms
NC
CHS_GND
R8
2.2kΩ
R9
2.2kΩ
HR911105A
SH1
SH2
U8
0603磁珠 120Ω@100MHz
3V3_SYS
A3V3_DP83825I
GND

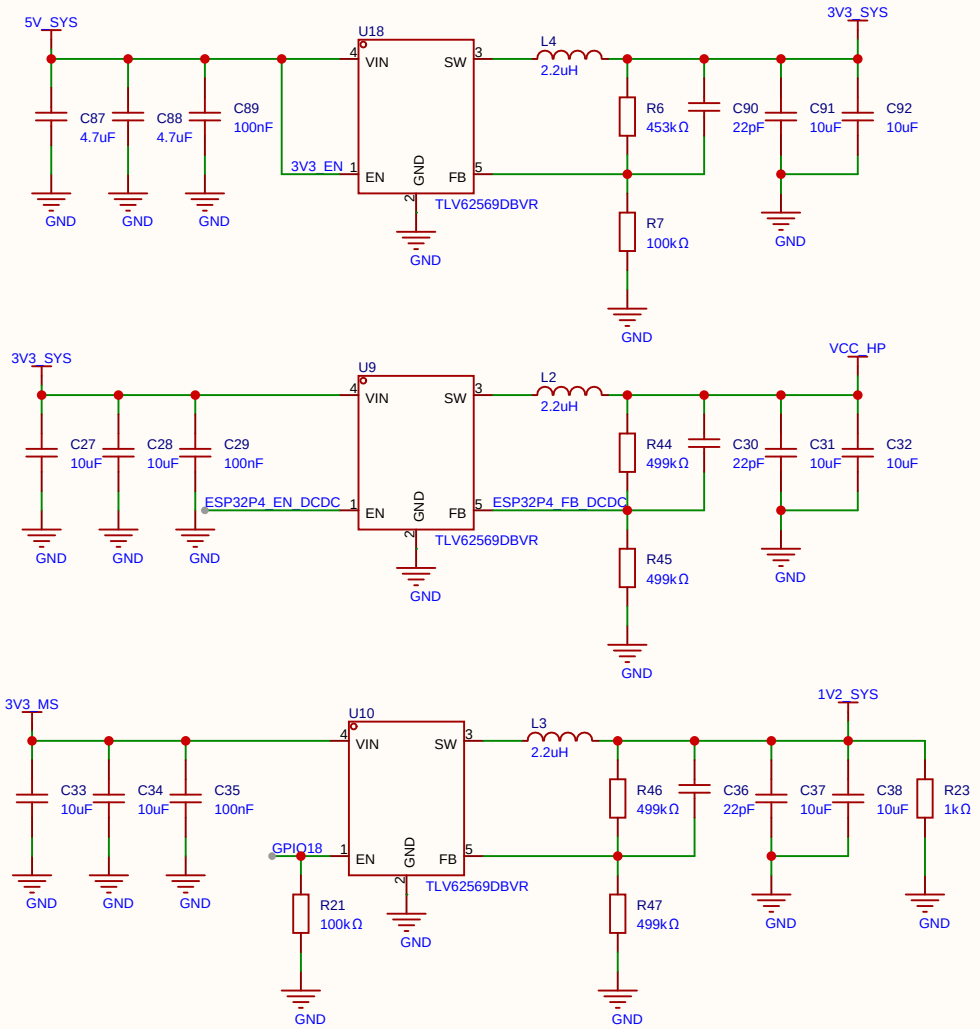

MS_SDA
MS_SCL
R61 0Ω
R31 0Ω
U13.1
ESP32-P4NRW32X
CRYSTAL
X2
40MHz
C15 22pF
C16 22pF
R26 10kΩ
U13.2ESP32-P4NRW32
VDD_HP_0/1/2/3: 每脚 100nF，附近再放 10uF
VDD_DCDCC: 10uF + 100nF
VDD_LDO: 10uF + 100nF
VDD_ANA: 1uF + 100nF
VDD_BAT: 1uF
VDD_USBPHY: 1uF + 100nF
VDDO_FLASH: 1uF + 100nF
VDDO_PSRAM: 1uF + 100nF
VDDO_3: 1uF
VDDO_4: 1uF，VDD_IO_5 旁边再放 10uF
FLASH
CSI/DSI
USB
USB_JTAG1
ZX-SH1.0-4PLT
U24
GD25Q127CYIGR
R25 10kΩ
R28 22Ω
R29 4.02kΩ
R30 4.02kΩ
C14 100nF
U22
ZX-SH1.0-2PLT
R48 22Ω
U4
GD5F2GQ5UEYIGR
R15 10kΩ
R19 10kΩ
C2 100nF
C109 10uF
C43 10uF
C59 1uF
C61 1uF
C64 1uF
C66 10uF

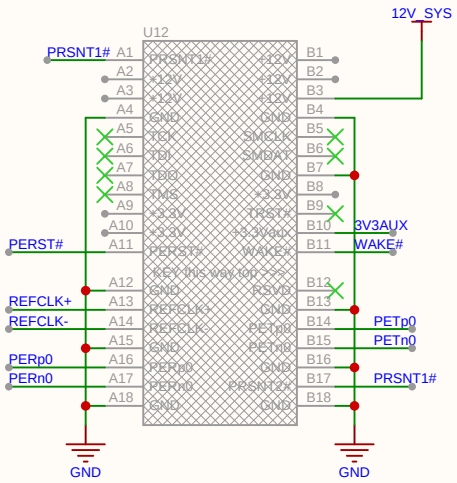

U6
MS2109
HXIN
HXOUT
REXT_MS
DP_MS
DM_MS
USB3V3_MS
MS_SCL
MS_SDA
3V3_MS
HDMIHPD
HDMIDET
1V2_SYS
HDMISDA
HDMISCL
A3V3_MS
XTIN
XTOUT
REXT
USBDP
USBDM
V33USB
I2SIN_SCLK
I2SIN_WS
I2SIN_SD0
I2S_MCLKOUT
GPIO2(SCL)
GPIO3(SDA)
DVDD33
HDMIRXHPD
HDMIRXDET
DVDD12
RXDDCSDA
RXDDCSCL
AVDD33
EP
VSS
AVDD33
MCU_SEL
DVDD33
SPDIFOUT
GPIO5
HDMIRX2P
HDMIRX2N
AVDD12
HDMIRX1P
HDMIRX1N
HDMIRX0P
HDMIRX0N
HDMIRXCP
HDMIRXCN
1V2_SYS
GPIO1_MS
LED_MS
A1V2_MS
HDMI_1+
HDMI_1-
HDMI_2+
HDMI_2-
HDMI_0+
HDMI_0-
HDMI_C+
HDMI_C-
GND
X3
24MHz
C17
C18
22pF
REXT_MS
R32
4.7kΩ
D2
D3
TPD4E05U06DQAR
HDMI1
HDMI-001S
TMDS Data 2+
TMDS Data 2-
TMDS Data 1 Shield
TMDS Data 0+
TMDS Data 0-
TMDS Data 0 Shield
TMDS Clock Shield
TMDS Clock+
TMDS Clock-
CEC
SCL
DDC/CEC Ground
Hot Plug Detect
Reserved (N.C.)
SDA
+5V Power
HDMISCL
HDMISDA
HDMIDET
HDMI_5V
R38
47kΩ
R39
47kΩ
R40
1kΩ
U21
0603磁珠 120Ω@100MHz
C20
100nF
U5
AT24C16C-STUM-T
SDA
SCL
Vcc
WP
C19
100nF
R34
2.2kΩ
MS_LED1
LED_MS
U7
TPS22918DBVR
VIN
VOUT
GPIO13
ON
QOD
CT
NET6
NET7
R20
22Ω
R22
100kΩ
C13
1nF
GPIO16
R35
4.7kΩ
MS_SCL
R36
4.7kΩ
MS_SDA
R37
4.7kΩ
U14
U15
U16
0603磁珠 120Ω@100MHz
C68
4.7uF
C69
10uF
C70
4.7uF
C71
C72
C73
C74
C75
C76
C77
C78
C79
C80
C81
C82
100nF
C3
C10
C12
C11
NC

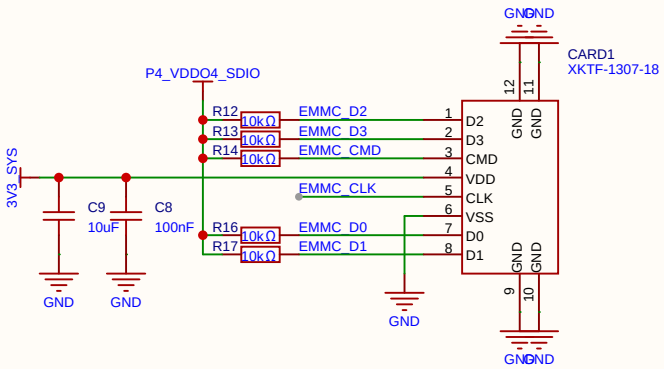

U1
CH343P
VIO
GND
VDD5
TXD
DTRTNOW
DCD
ACT#
VBUS
RI
CTS
DSR
RTS
EP
RXD
V3
UD+
UD-
3V3_SYS
5V_SYS
GND
C39
100nF
C40
100nF
C41
100nF
C42
100nF
CH343P0_USB_TXD
CH343P0_USB_RTS
CH343P0_USB_DTR
CH343P0_USB_RXD
CH343P0_USB_D+
CH343P0_USB_D-
U2
EMH4T2R
CHIP_PU
C1
1uF
R1
10kΩ
R3
2.2kΩ
RMII_TXD1_GPIO35
USB2
HX TYPE-C 16P L8.35
A1B12
A4B9
B8
A5
B7
A6
B6
A8
B5
B4A9
B1A12
R2
5.1kΩ
R4
5.1kΩ
R5
5.1kΩ
LED1
NCD0603R1
C4
NC
U20
0603磁珠 120Ω@100MHz
5V_USB
U17
CH343P
C84
100nF
C85
100nF
C86
100nF
GPIO50_USB2_TXD
GPIO51_USB2_RXD
USB_VBUS
USB2_D+
USB2_D-
H1
X6521WR-2x12H-C60D30
USB_VBUS0
DM_HID
DP_HID
CUT THIS PIN
RST_SW2
RST_SW1
PWR_SW2
PWR_SW1
GPIO2
GPIO3
MS_SDA
MS_SCL
U11
AQS221R2S
GPIO4
GPIO5
R49
510Ω
R50
510Ω
R51
510Ω
R52
10kΩ
R53
2.2kΩ
R54
10kΩ
3V3AUX
12V_SYS
GPIO1
GPIO0

3V3
1V2

VCCHP




U12
12V_SYS
PRSNT1#
A1
PRSNT1#
+12V
B1
A2
+12V
+12V
B2
A3
+12V
+12V
B3
A4
GND
GND
B4
A5
TCK
SMCLK
B5
A6
TDI
SMDAT
B6
A7
TDO
GND
B7
A8
TMS
+3.3V
B8
A9
+3.3V
TRST#
B9
A10
+3.3V
+3.3VAUX
B10
3V3AUX
PERST#
A11
PERST#
WAKE#
B11
WAKE#
KEY (NO PIN 100 >>)
A12
GND
PSVD
B12
REFCLK+
A13
REFCLK+
GND
B13
REFCLK-
A14
REFCLK-
PETp0
B14
PETp0
A15
GND
PETn0
B15
PETn0
PERp0
A16
PERp0
GND
B16
PERn0
A17
PERn0
PRSNT2#
B17
PRSNT1#
A18
GND
GND
B18
GND
GND


P4_VDDO4_SDIO
3V3_SYS
R12
R13
R14
R16
R17
10kΩ
EMMC_D2
EMMC_D3
EMMC_CMD
EMMC_CLK
EMMC_D0
EMMC_D1
C9
10uF
C8
100nF
GND
CARD1
XKTF-1307-18
D2
D3
CMD
VDD
CLK
VSS
D0
D1
1
2
3
4
5
6
7
8
9
10
11
12